GAURAV GUPTA

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PROFESSIONAL SUMMARY

Software engineer with 2 years building C++ and Python services on AWS and Oracle Cloud for enterprise financial software at Tally Solutions, progressing from intern to SDE-II. Shipped an automated failover and self-healing system that restored 30+ client systems during an outage, and rewrote 20+ AWS Lambda functions as platform-agnostic services for a zero-loss migration to Oracle Cloud. MSc in Computer Science with Distinction from the University of St Andrews, focused on Distributed Systems, Data-Intensive Systems and Critical Systems Engineering.

SKILLS

BACKEND ENGINEERING | C++, Python, FastAPI, REST APIs, PostgreSQL, Redis, MongoDB, SQL, Alembic schema migrations
CLOUD & DEVOPS | AWS (Lambda, EC2, S3), Oracle Cloud (OCI), Docker, Docker Compose, CI/CD, GitHub Actions.
AI ENGINEERING | LangGraph agents, LLM integration and structured extraction, Gemini API, Ollama, Claude Code.
TESTING & TOOLING | pytest, unit testing, LLM mocking, Git, GitHub, Postman, Jira, Confluence, JavaScript, TypeScript, React

PROFESSIONAL EXPERIENCE

TALLY SOLUTIONS PVT. LTD. - BENGALURU, INDIA | June 2023 - June 2025

Software Development Engineer II | C++, Python, AWS, Oracle Cloud (OCI)

- Took ownership of an **unassigned, customer-impacting reliability issue**, analysed it with the **AWS and CloudOps teams**, then designed and shipped an **automated failover and self-healing system end-to-end**, which **restored 30+ customer systems during an AWS region outage** and earned the company's annual **Customer Delight Champion Award**.
- Migrated **20+ production AWS Lambda functions across 10+ components** from **AWS** to **Oracle Cloud**, redesigning them as **provider-agnostic** functions to remove future vendor lock-in.
- Improved core product stability by resolving critical defects in **XML and Math Parsers** used for financial calculations in Tally migration software, enabling enterprise customers to update local desktop apps **without data loss**.
- **Reviewed code across multiple teams** and **mentored new joiners** as Process, Methodology and Tools trainer, giving feedback on code quality and bugs. Built **TDL placeholder components** that let **third-party vendors extend the Tally UI**.

EDUCATION

MSc., Computer Science (16.5/20, **Distinction, Dean's List**) - University of St Andrews, UK | Sept 2025 - Sept 2026

B.Tech., Computer Science (8.88/10, **Distinction**) - Dayananda Sagar University, Bengaluru, India | Aug 2019 - June 2023

PROJECTS

PRIMAROS: DRUG INTERACTION PLATFORM | MSC DISSERTATION | Python, FastAPI, PostgreSQL, Redis

- Architected a **layered clinical decision-support system** integrating three heterogeneous medical data sources (DrugBank, OpenFDA, RxNorm) into a unified domain model of **21,500+ drugs** and **2.9 million interactions**. Built streaming **ingestion pipelines** capable of processing **1.8 GB DrugBank XML and 8 GB OpenFDA label sets** while keeping memory usage independent of input file size.
- Built a **REST API backend** supporting **trigram-indexed fuzzy drug search** over the ingested data. Added **Redis caching** for recently searched drug interactions to reduce retrieval latency, integrated **Alembic-based database schema migrations**, and supported containerised deployment using **Docker Compose**.

INTERVIEW PREPARATION AGENT | Python, LangGraph, Gemini API, Tavily API | GitHub

- Built a stateful **LangGraph agent** with three modes: CS course revision, learning, and behavioural coaching. "Revision" dynamically adapts question difficulty using **per-topic confidence scores** persisted in SQLite, while "Coaching" uses **structured LLM extraction** to score responses and provide **STAR-format** feedback. "Learning" tracks competency coverage across sessions, with question banks updated via **Tavily web search**.

NIMBLE-TRIAGE: OPEN-SOURCE LOG TRIAGE CLI | Python, Ollama, SQLite | GitHub | PyPI

- Shipped a **Python CLI on PyPI** that uses a **local LLM** to flag which log entries need human attention, with **cached decisions** and **capped model requests** to keep cost and behaviour predictable.

EXTRACURRICULARS AND VOLUNTEERING

- Process, Methodology, and Tools (PMT) trainer and Hackathon Mentor for 10 teams (2024). — Tally Solutions Pvt. Ltd.
- C Language Trainer for 70+ students and pre-placement session lead. — DSU (University)